

A139 · Mapping Cis-Regulatory Programs of Pancreatic β Cells in Health and Diabetes

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Session: Day 2 · Fri Oct 2, 2026 · 2:15–4:15 PM

Pancreatic β -cell dysfunction is central to both type 1 (T1D) and type 2 diabetes (T2D), yet the regulatory programs underlying β -cell identity and failure remain incompletely characterized. The scarcity of primary human pancreatic tissue has further limited our ability to define β -cell and other endocrine cell states. Despite numerous efforts to profile chromatin accessibility in human islets, these datasets remain fragmented across studies and biological contexts, limiting a systematic view of the pancreatic regulatory landscape.

To address this gap, we systematically collected and uniformly processed over 20 publicly available bulk and single-cell chromatin accessibility datasets comprising more than 250 human donors. Using a standardized framework based on ENCODE protocols, we constructed a unified map of candidate cis-regulatory elements (cCREs) across pancreatic endocrine (β , α , and δ) and exocrine cell types. This resource spans healthy, prediabetic, T1D, and T2D states, as well as ex vivo perturbation models that mimic disease-relevant conditions, including inflammation and endoplasmic reticulum stress.

We expanded the pancreatic regulatory landscape at cell-type resolution, identifying thousands of cCREs selectively accessible in β cells across diabetic and stress conditions. Importantly, we recovered additional cCREs absent from the existing ENCODE registry, many selectively active in endocrine and disease-associated states, suggesting that these rare contexts remain underrepresented in existing references. Applying sequence-based deep learning models of chromatin accessibility, we further nominated regulatory sequence features and candidate transcription factors that distinguish β cells in disease-relevant states. Together, this resource provides a more complete picture of β -cell regulation in the context of diabetes.